

Use Case 1: Autoscaling and Load Balancing

Scenario: Create a scalable e-commerce website.

Tasks:

- o Set up an EC2 instance for hosting a simple website.
- o Configure an Auto Scaling group to handle traffic spikes.
- o Use an Elastic Load Balancer to distribute traffic across instances.
- o Monitor scaling events using CloudWatch.

Step 1: Set Up the EC2 Instance to Host the Website

1.1 Launch an EC2 Instance

1. **Log in to AWS Console:**
 - o Go to the **EC2 Dashboard** and click **Launch Instance**.
2. **Choose an AMI:**

- Select an **Amazon Machine Image (AMI)** for a simple web server. A good option is the **Amazon Linux 2 AMI** or **Ubuntu**.
3. **Choose an Instance Type:**
 - For this use case, start with a **t3.micro** or **t3.small** instance (this can be changed later based on your needs).
 4. **Configure Instance Details:**
 - Select your **VPC** and configure the **Subnet** (public subnet for web server).
 - Ensure the **Auto-assign Public IP** option is enabled (for public access).
 5. **Add Storage:**
 - Choose **8 GB** or more of **General Purpose SSD (gp2)** storage.
 6. **Configure Security Group:**
 - Create a **Security Group** that allows:
 - **HTTP (Port 80)** for web traffic.

- **SSH (Port 22)** for administrative access (only from trusted IPs).

7. **Launch the Instance:**

- Review and launch the instance. Make sure you select or create a key pair for SSH access.

1.2 Install Web Server Software

- Once the EC2 instance is launched, SSH into the instance and install a web server (e.g., **Apache** or **Nginx**).

Example (for Apache on Amazon Linux 2):

```
bash
```

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
echo "Welcome to My E-commerce Site!" |
```

```
sudo tee /var/www/html/index.html
```

- Access your website by navigating to the **Public IP address** of the EC2 instance in a web browser.
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Step 2: Set Up an Auto Scaling Group

2.1 Create a Launch Configuration

1. Go to the **EC2 Dashboard** and select **Auto Scaling** under **AUTO SCALING**.
2. Click **Create Auto Scaling group**.
3. **Create a Launch Configuration:**
 - Choose the AMI (same as the EC2 instance you created).
 - Select the **instance type** (e.g., **t3.micro**).
 - Set up storage (the same as the EC2 instance).
 - Select a **Security Group** (same as the EC2 instance).
 - Set up the **key pair** for SSH access.
4. Click **Create Launch Configuration**.

2.2 Create an Auto Scaling Group

1. After creating the launch configuration, create an **Auto Scaling group**:
 - Choose the **VPC** and **subnets** (select public subnets for load-balanced web instances).
 - Set a **desired capacity** (e.g., **2 instances**).
 - Define **minimum** and **maximum capacity** (e.g., **1** and **5** instances).
2. **Configure Scaling Policies**:
 - Set up **Target Tracking Scaling** or **Step Scaling** policies:
 - Example: **Scale up** when CPU usage > 70%, and **Scale down** when CPU usage < 30%.
3. Click **Create Auto Scaling Group**.

Step 3: Set Up Elastic Load Balancer (ELB)

3.1 Create an Application Load Balancer (ALB)

1. Go to the **EC2 Dashboard > Load Balancers** and click **Create Load Balancer**.
2. Choose **Application Load Balancer (ALB)**.
3. Configure the **Load Balancer**:
 - Name the load balancer (e.g., **Ecommerce-ALB**).
 - Select the **VPC** and **subnets** (choose the same subnets as the EC2 instance for high availability).
 - Choose **internet-facing** for public access.
 - Configure **Security Groups** (allow HTTP (Port 80)).
4. **Configure Listeners and Routing**:
 - Set **HTTP** (Port 80) as the listener.
 - Create a target group to register EC2 instances with the ALB.

5. Click **Create**.

3.2 Register Instances with the Load Balancer

1. Go to the **Target Groups** section and select the target group you created.
 2. Under the **Targets** tab, click **Register Targets**.
 3. Select the EC2 instances (from the Auto Scaling group) that will be load-balanced by the ALB.
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Step 4: Configure CloudWatch Monitoring and Auto Scaling Triggers

4.1 Monitor EC2 Instances and Auto Scaling

1. Go to **CloudWatch** in the AWS Console.
2. Set up **Alarms** to monitor metrics such as **CPU utilization**, **memory usage**, or **network traffic**.
 - Example: Create an alarm that triggers when CPU utilization exceeds **70%** for **5 minutes**.

- Action: Trigger an Auto Scaling policy to **add a new instance**.

4.2 Set Up Auto Scaling Policies Using CloudWatch Alarms

1. In the Auto Scaling group, link the CloudWatch alarms to trigger scaling events.
 - Example:
 - **Scale Out:** Add an instance if CPU utilization > 70% for 5 minutes.
 - **Scale In:** Remove an instance if CPU utilization < 30% for 5 minutes.
2. **Test Scaling Events:**
 - Manually stress test the application (e.g., using **Apache Benchmark** or **ab**) to trigger scaling events.
 - Monitor **CloudWatch** for scaling events (instances added or removed).

Step 5: Test the E-Commerce Website

1. **Simulate Traffic:**

- Simulate real traffic by visiting the website multiple times.
- **CloudWatch** will monitor the traffic and trigger scaling policies to add or remove EC2 instances based on demand.

2. **Monitor Load Balancer:**

- Use the **Application Load Balancer's** dashboard to monitor traffic distribution across your EC2 instances.